

# Recognizing multiple needle insertion attempts for performance assessment in central venous catheterization training

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## ABSTRACT

**Purpose:** Computer-assisted skill assessment has traditionally been focused on general metrics related to tool motion and usage time. While these metrics are important for an overall evaluation of skill, they do not address critical errors made during the procedure. This study examines the effectiveness of utilizing object detection to quantify the critical error of making multiple needle insertion attempts in central venous catheterization. **Methods:** 6860 images were annotated with ground truth bounding boxes around the syringe attached to the needle. The images were registered using the location of the phantom, and the bounding boxes from the training set were used to identify the regions where the needle was most likely inserting the phantom. A Faster region-based convolutional neural network was trained to identify the syringe and produce the bounding box location for images in the test set. A needle insertion attempt began when the location of the predicted bounding box fell within the identified insertion region. To evaluate this method, we compared the computed number of insertions to the number of insertions identified by human reviewers. **Results:** The object detection network had an overall mean average precision (mAP) of 0.71. This tracking method computed an average of 4.40 insertion attempts per recording compared to a reviewer count of 1.39 attempts per recording. **Conclusions:** The difference in the number of insertion attempts identified by the computer and reviewers decreases with an increasing mAP, making this method suitable for detecting multiple needle insertions using an object detection network with a high accuracy.

**Keywords:** Object detection, computer-assisted skill assessment, central venous catheterization

## 1. INTRODUCTION

Effective feedback is a critical component of competency-based medical education; however, residents are often not satisfied with the quality and frequency of feedback they receive during training. One of the most prevalent reasons for this lack of satisfying feedback is the low availability of supervisors. In the emergency department, 83.7% of attending physicians report that time constraints are a barrier to providing quality feedback for residents [1]. Computer-assisted skill assessment is an effective solution to this issue, as it provides trainees with the real-time comprehensive feedback that many physicians do not have the time to give. It is necessary that residents practice procedures such as central venous catheterization (CVC) regularly, as they are commonly performed for a wide range of patients and conditions. It is also likely that residents will have to regularly transfer the skills they acquired in training for CVC to real life situations. In these cases, computer-assisted skill assessment is an especially valuable tool, as it can provide consistent feedback during the multiple times a trainee would be performing the procedure. Central Line Tutor, developed by Hisey et al., is a computer-based platform used for skill assessment in CVC that provides real-time feedback and instruction without requiring continuous expert supervision [2]. Using this platform to aid in training for CVC would give trainees the opportunity to receive more frequent quality feedback without creating a significant draw on time for supervisors.

Traditionally, computer-based skill assessment platforms like Central Line Tutor use tracking methods that rely on external tracking systems such as electromagnetic (EM) or optical tracking to evaluate general metrics such as tool motion and usage time [3,4]. Though these methods have been effective in evaluating trainee skill, using such additional tracking systems can be very costly and cumbersome. External tracking systems also make it more difficult to transfer simulation-based training to the real world, as external trackers require calibration and physically alter the tools. As a result, video-based tracking systems are becoming more prevalent in order to perform more cost-effective skill assessments using surgical tools that are easier to maneuver. Preliminary success using tracking methods such as object detection to compute similar motion-based metrics was shown using video data [5]. In one such method, O'Driscoll et al. trained a faster region-

based convolutional neural network (Faster R-CNN) to assess trainee skill in CVC using 2D tool path length and usage time. They compared the tracking ability of Faster R-CNN to that of EM trackers used to track the same trainee assessment metrics. A strong correlation was found between the effectiveness of the use of Faster R-CNN and EM tracking, making the use of object detection a suitable method for evaluating tool motion for computer-based skill assessment [6].

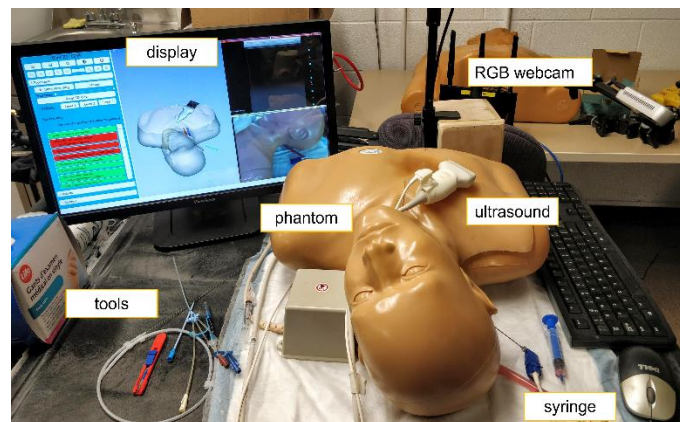
The motion-based metrics computed in these previous systems are good measures of trainee performance, however they are also very generalized across many different procedures. It is necessary that, in the development of a computer-assisted assessment platform for an individual procedure, the system also monitors for procedure specific errors in addition to those general metrics. Some platforms, namely Central Line Tutor, have taken steps towards addressing some of these errors. The Central Line Tutor platform includes prompts for upcoming tasks and visual cues alerting trainees of workflow errors specific to CVC. However, Central Line Tutor is still lacking metrics for informing trainees of some major critical errors of the procedure [7]. One such critical error is the number of needle insertion attempts made by the trainee. For CVC, the chance of experiencing mechanical complication is more than eight times greater when cannulation is attempted more than once [8]. Including the number needle insertion attempts into the list of standard computed metrics used for skill assessment in CVC is important. This would ensure that platforms such as Central Line Tutor provide trainees with sufficient guidance to improve the overall outcome of the procedure.

In this study, we aim to track the number of needle insertions attempted during CVC using webcam video data. We hope to achieve this by using video-based tracking. Specifically, we expand on the method developed by O'Driscoll et al. that uses object detection with Faster R-CNN to recognize surgical tools. This will build upon the existing applications that object detection has in evaluating general trainee skill, allowing us to provide a more comprehensive evaluation of trainee performance in CVC using additional procedure specific critical error metrics.

## 2. METHODS

### 2.1 Data acquisition

Video recordings were obtained from five anesthesiologists certified to practice without supervision and one medical student who had never performed CVC before. They were recorded performing CVC on the Central Line Tutor setup five times each, amounting to a total of 30 recordings. The recordings were taken using an RGB webcam mounted to a tripod from a sideview. This allowed for a clear view of the procedure to be obtained while preserving both the anonymity of the participant and their ability to perform CVC unobstructed. The Central Line Tutor setup used to record the data is pictured in Figure 1.



**Figure 1.** The setup for Central Line Tutor used to record the video data.

The videos were broken down into individual frames using the Deep Learn Live extension of 3D Slicer (<https://github.com/SlicerIGT/aigt/tree/master/DeepLearnLive>). 3D Slicer ([www.slicer.org](http://www.slicer.org)) is a free, open-source software platform designed for medical image analysis and scientific visualization. Each of the frames was annotated with a bounding box location of the syringe and the phantom. The frames were also annotated with the time that the needle initially intersects the phantom and the time it was fully removed from the phantom for each needle insertion attempt. The number of needle insertion attempts was identified for each video by 3 reviewers.

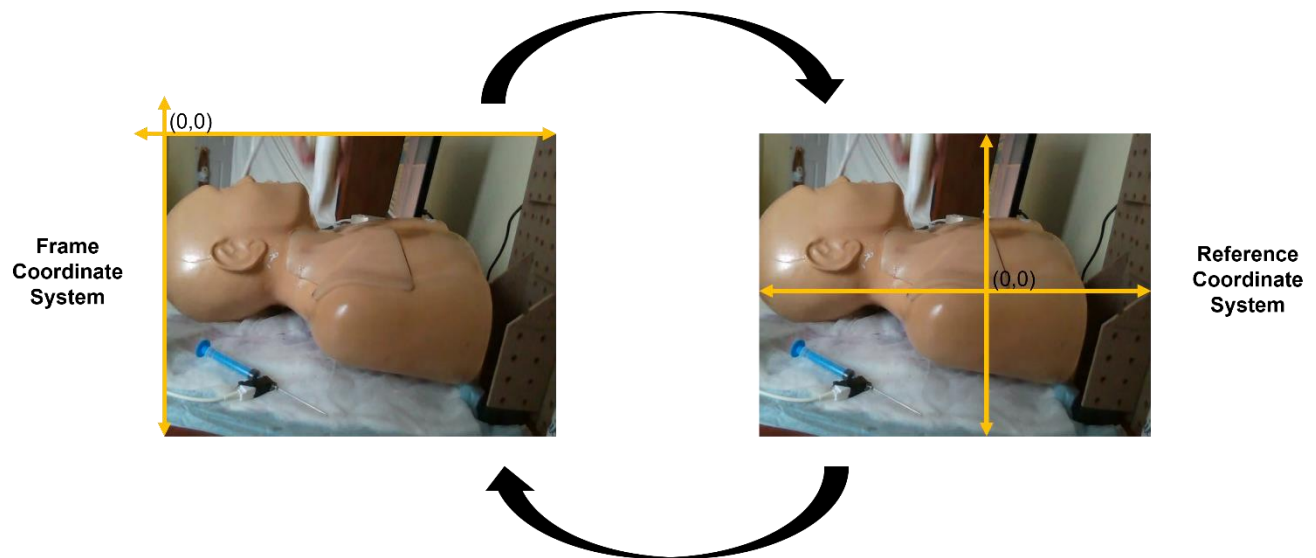
## 2.2 Image registration

As the data was taken from multiple participants at different times, there is some variance in the focus and angle of videos that occurs due to differences in the camera position during the different recordings. To ensure the reliability of our computed results despite those varying camera positions, we registered all the frames using the phantom bounding boxes. This allowed all bounding box coordinates to be expressed using a single coordinate system. The location of the phantom was used to define a reference coordinate system that the syringe coordinates could be transformed to. For the coordinate system of a frame, the origin corresponds to the top left corner of the image, whereas for the reference coordinate system, all frame coordinates are transformed such that the origin corresponds to the center of the phantom bounding box. The area used when registering the frames was determined using object detection on the phantom to create bounding boxes around the phantom's borders. An example of the borders created by the bounding box in comparison to the full frame is shown in Figure 2.



**Figure 2.** An example of the area used in the frame registration created by the ground truth phantom bounding box.

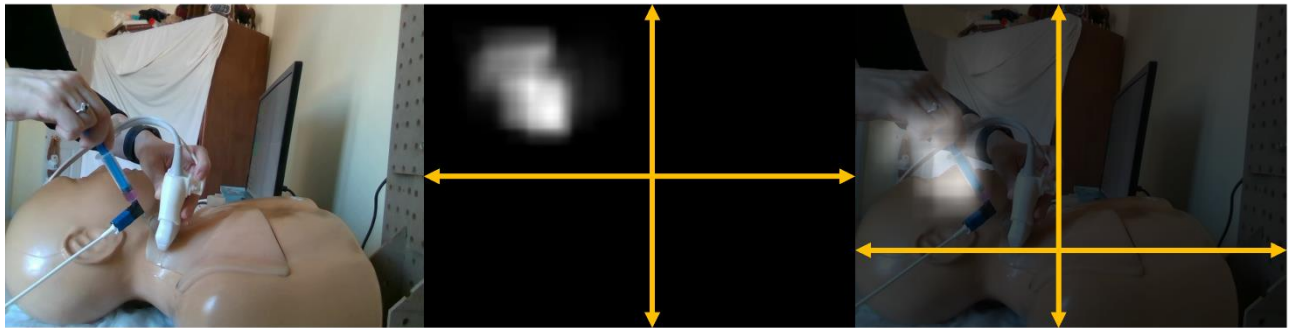
The conversion between these two coordinate systems is illustrated in Figure 3. The reference coordinate system is used to create the heat map and determine confidence thresholds, and the frame's coordinate system is used to apply the heat map to each video and determine the confidence value equated to a needle in a frame.



**Figure 3.** An illustration of the conversion from the coordinate system from the frame coordinate system to the reference coordinate system based on the location of the phantom.

## 2.3 Heat map construction

We constructed a heat map using the registered frames to determine the regions where the probability of a needle insertion attempt occurring was the highest. The heat map was constructed from the locations of the ground truth bounding box annotations for the syringe. The coordinates of the syringe were transformed with respect to the ground truth bounding box annotations for the phantom. To determine the confidence value for each pixel, we calculated the number of bounding boxes where the needle tip was intersecting the phantom that overlapped with that pixel. We only used the bounding boxes from the training set of bounding boxes to construct the heat map. We then normalized the number of overlapping bounding boxes by dividing the value for each pixel by the maximum value to create confidence values ranging between 0 and 1. The confidence values were then stored for each pixel as an image that visualized the values as a grayscale. An example of a confidence heat map can be seen in Figure 4, where the lightest areas illustrate the areas of the frame where we are most confident a needle attempt could be occurring, and the darkest areas show the areas of least confidence. A version of this heatmap overlaid with a frame from one of the recordings has also been included to show where these areas of highest and lowest confidence occur in relation to the phantom.



**Figure 4.** *Left.* Original image. *Center.* Confidence heat map centered around the middle of the frame. *Right.* Image overlaid with the transformed heat map (centered around the origin of the heat map).

## 2.4 Confidence threshold computation

Once the confidence heat map was created, we computed upper and lower thresholds that define a needle insertion attempt start and end respectively. For each frame we obtained the confidence value for the pixel that corresponded with the center of the ground truth bounding box identifying the syringe. We then used the timestamps given by reviewers that denoted when a needle insertion attempt was beginning to define the confidence values used for the upper threshold, and the timestamps that denoted when the attempt had ended to define the confidence values used for the lower one. We then averaged both sets of confidence values to obtain a single value for each threshold. These thresholds were computed based only on the images used to train the object detection network.

## 2.5 Object detection

To count needle insertion attempts for novel data, we trained an object detection network, Faster R-CNN, to recognize both the syringe and the phantom. The network was trained using the ground truth bounding boxes. The network was trained using a leave-two-user-out cross-validation scheme. The videos from two participants were used to make up the test set for each fold, with one video from each of the remaining participants becoming the validation set and the remaining 16 videos becoming the training set. The training set was balanced such that the data used included equal instances of the syringe and the phantom bounding boxes. In total, there were 6860 frames annotated with ground truth bounding boxes surrounding the syringe and the phantom that were used to train the network. To evaluate the performance of the network, we computed the network's mean average precision (mAP) as well as the mAP for each class.

## 2.6 Detection of insertion attempts

We computed the number of needle insertion attempts by calculating the number of times the system observed a needle insertion attempt start and end. We defined the start of a needle insertion attempt as when the confidence value which corresponded to the pixel at the center of the syringe bounding box was higher than the upper threshold. The end of a needle insertion attempt was defined as occurring when the confidence value corresponding to the center of the syringe bounding box fell below the lower threshold. To evaluate the efficacy of our method for tracking needle insertion attempts,

we computed the number of needle insertion attempts from the predicted syringe location data and compared the results to the average number of needle insertion attempts identified in each video by 3 reviewers.

### 3. RESULTS

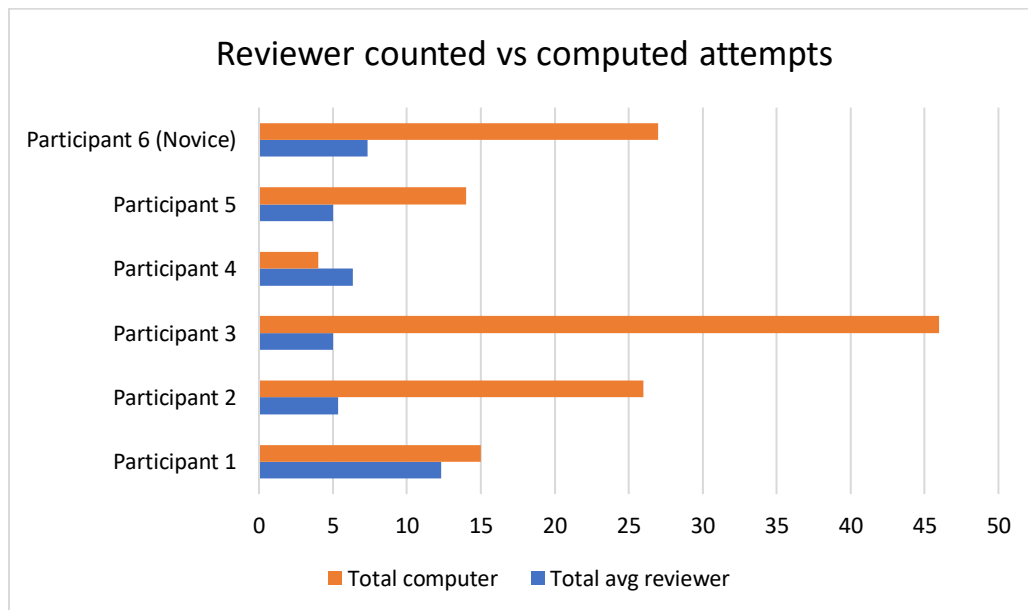
The heat maps and confidence values for the syringe locations were computed from the training data to determine the thresholds that define a needle insertion attempt. The upper threshold that defines when a needle insertion begins was 0.65, and the lower threshold was 0.33.

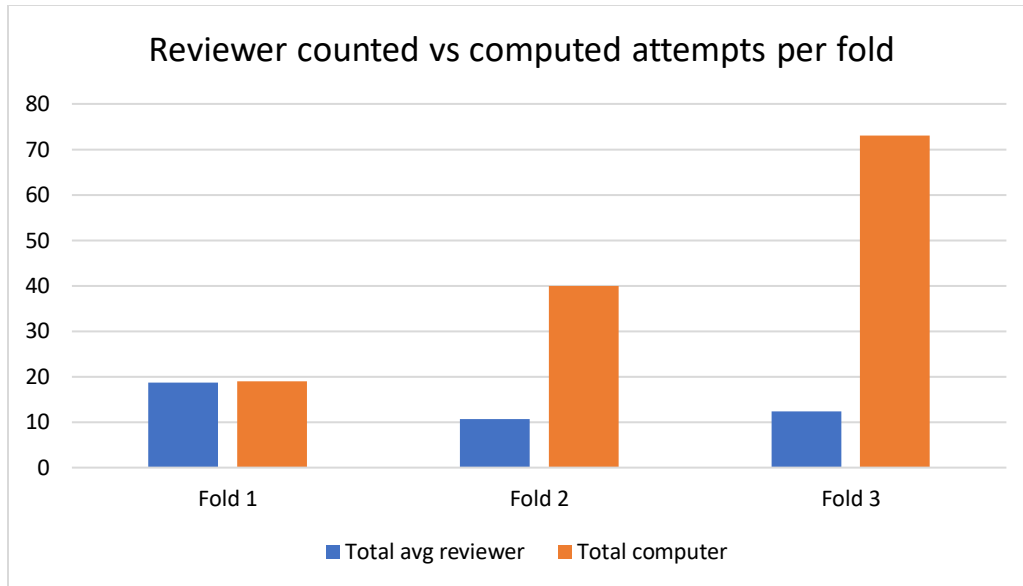
The overall mAP for the network was 0.714. Fold 1 had the highest overall mAP of 0.832 and syringe mAP of 0.665. The mAP for the phantom was consistent, with a variation of 0.001 between folds. Table 1 shows the mAP for the syringe, phantom, and network overall for each fold.

**Table 1.** The mean average precision for each fold

|          | <b>syringe</b> | <b>phantom</b> | <b>overall</b> |
|----------|----------------|----------------|----------------|
| <b>1</b> | 0.665          | 0.999          | 0.832          |
| <b>2</b> | 0.431          | 0.999          | 0.715          |
| <b>3</b> | 0.191          | 1.000          | 0.595          |

The overall average number of computed needle insertion attempts per recording was 4.40 compared to the 1.39 per recording count averaged from the reviewers. Participants 1 and 4 had the smallest differences between insertion attempts per recording, with both participants having a difference of  $>0.6$  between the computed and reviewer counted insertion attempts. Fold 1, which the results for participants 1 and 4 were derived from, was the fold with the smallest difference between computed and reviewer counted attempts. Figure 5 displays the number of insertions counted by the computer in comparison to a reviewer count for the test data for each participant and each fold.





**Figure 5.** *Top.* A comparison between the reviewer and computer count for the number of needle insertion attempts for each participant. *Bottom.* A comparison between the reviewer and computer count of the insertion attempts per fold.

## NEW OR BREAKTHROUGH WORK TO BE PRESENTED

The aim of this study was to expand upon the existing applications that object detection has in evaluating general trainee skill to develop a metric specifically for evaluating skill in performing CVC based on critical errors. It achieves this by presenting a method for computing the number of needle insertion attempts made during CVC using only a side view video of the procedure. We demonstrate that the number of needle insertion attempts can be reliably computed using video-based tracking methods that have previously shown promise in measuring general metrics such as tool motion. This shows that object detection is a promising method for computing a procedure-specific critical error metric for CVC in addition to the general metrics it has already shown promise in computing.

## DISCUSSION AND CONCLUSIONS

The number of attempts computed using the video-based method was very similar to the number of attempts identified by a human reviewer for participants 1 and 4. The results for both participants came from data trained in the first fold. The first fold also had the highest mAP overall and for the syringe. This trend of having a smaller difference in attempt counts with an increasing mAP can also be seen with the other folds. This indicates that, with the use of an object detection network with a high accuracy, this method can compute a similar number of needle insertion attempts to a reviewer. This makes the approach a promising approach to recognizing multiple needle insertion attempts. A potential cause for this higher accuracy for the first fold is the number of images that were used to train each fold. The data from participants 5 and 6 contained a greater number of images compared to the remaining 4 participants. Since the data from both participants was used to train fold 1, compared to only one participant's data being present in the training set of the other two folds, the first fold was trained with a larger number of images compared to the other two folds.

One of the factors that limited us in our experiments was the small size of the dataset, which was made up of mostly expert participants. In future work, we will expand upon the dataset, especially focusing on increasing the number of novice participants. With this expanded dataset, we aim to be able to make a comparison between the number of needle insertion attempts made by experts and the number made by novices and evaluate how the number of insertion attempts relate to skill. We also hope that a larger dataset that uses more frames to train each fold will improve the performance of our object detection network.

We also hope to support a greater range of camera positions in future datasets. In the current dataset, the range of camera positions was very small, as the camera used was mounted on a static arm that experienced very little movement while collecting the dataset. This means that we can only evaluate whether our method of image registration is effective over a

small range. To find the true range of side views that this method will work within, we will include recordings taken from a greater range of camera positions and compare the effectiveness of our algorithm in computing the number of insertion attempts for these different positions. In addition to using a larger variety of camera positions for the sideview recordings, we plan to incorporate recordings using different types of cameras into the dataset, namely a camera recording from an above view and two depth cameras recording from both the above and side views. We aim to see if adding these cameras into our method will increase the accuracy of both the computed number of needle insertion attempts and the predicted start and end times of the insertion attempts. In the future this work will be integrated into the is integrate this system into Central Line Tutor system to ensure the skill assessment metrics it computes are inclusive of both general indicators of skill assessment and critical errors specifically related to CVC.

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## REFERENCES

- [1] Yarris, L. M., Linden, J. A., Gene Hern, H., Lefebvre, C., Nestler, D. M., Fu, R., Choo, E., LaMantia, J., Brunett, P., "Attending and resident satisfaction with feedback in the emergency department," *Academic Emergency Medicine* **16** (2009).
- [2] Hisey R, Camire D, Erb J, Howes, Fichtinger G, Ungi T. System for central venous catheterization training using computer vision-based workflow feedback. *IEEE Transactions on Biomedical Eng.* 2021 Nov. 2 [3] Harada, K., Morita, A., Minakawa, Y., Baek, Y. M., Sora, S., Sugita, N., Kimura, T., Tanikawa, R., Ishikawa, T., et al., "Assessing Microneurosurgical skill with MEDICO-ENGINEERING TECHNOLOGY," *World Neurosurgery* **84**(4), 964–971 (2015).
- [4] Holden, M. S., Xia, S., Lia, H., Keri, Z., Bell, C., Patterson, L., Ungi, T., Fichtinger, G., "Machine learning methods for automated technical skills assessment with instructional feedback in ultrasound-guided interventions," *International Journal of Computer Assisted Radiology and Surgery* **14**(11), 1993–2003 (2019).
- [5] Pérez-Escamirosa, F., Chousleb-Kalach, A., Hernández-Baro, M. del., Sánchez-Margallo, J. A., Lorias-Espinoza, D., Minor-Martínez, A., "Construct validity of a video-tracking system based on orthogonal cameras approach for objective assessment of laparoscopic skills," *International Journal of Computer Assisted Radiology and Surgery* **11**(12), 2283–2293 (2016).
- [6] O'Driscoll, O., Hisey, R., Holden, M., Camire, D., Erb, J., Howes, D., Ungi, T., Fichtinger, G., "Feasibility of object detection for skill assessment in central venous catheterization," *Medical Imaging 2022: Image-Guided Procedures, Robotic Interventions, and Modeling* (2022).
- [7] Hisey, R., Camire, D., Erb, J., Howes, D., Fichtinger, G., Ungi, T., "System for central venous catheterization training using computer vision-based workflow feedback," *IEEE Transactions on Biomedical Engineering* (2021).
- [8] Schummer, W., Schummer, C., Rose, N., Niesen, W.-D., Sakka, S. G., "Mechanical complications and malpositions of central venous cannulations by experienced operators," *Intensive Care Medicine* **33**(6), 1055–1059 (2007).